

Lise Le Boudec

PhD Student at ISIR, Sorbonne University

ACADEMIC BACKGROUND

2022-now

PhD Student

PHYSICS BASED DEEP LEARNING: LEARNING PDE SOLVERS · ISIR (MLIA)



PhD Student in the MLIA team at ISIR lab (Sorbonne University). Publications:

- Operator Learning with Neural Fields: Tackling PDEs on General Geometries. Louis Serrano, **Lise Le Boudec**, Armand Kassai Koupai, Thomas X Wang, Yuan Yin, Jean-Noël Vittaut, Patrick Gallinari. **NeurIPS 2023 - Poster**.
- Learning a Neural Solver for Parametric PDE to Enhance Physics-Informed Methods. **Lise Le Boudec**, Emmanuel de Bezenac, Louis Serrano, Ramon Daniel Regueiro-Espino, Yuan Yin, Patrick Gallinari. **ICLR 2025 - Poster**.
- ENMA: Tokenwise Autoregression for Continuous Neural PDE Operators. Armand Kassai Koupai, **Lise Le Boudec**, Louis Serrano, Patrick Gallinari. **NeurIPS 2025 - Spotlight Poster**.
- EEfficient Generative Transformer Operators For Million-Point PDEs. Armand Kassai Koupai, **Lise Le Boudec**, Patrick Gallinari, 2025 - Preprint.



2022

Master 2 Mathematics

LEARNING AND ALGORITHMS · Sorbonne université



statistical learning, deep learning



2021

Master in Management and Business Administration: Entrepreneurship and Management of Innovative Projects

LEARNING AND ALGORITHMS · IGR-IAE Rennes



Finance, Marketing, Management, Entrepreneurship



2021

Ingeneering Shool

APPLIED MATHEMATICS · INSA Rennes



- 2021: Optimal control course & Double-diploma INSA-IGR
- 2020 : Semester at UQAM in Montreal (Canada)
- 2019: Introduction to research: « Discovery of new physical laws through parsimonious representations»
- 2018: Mathematical Engineering Department (Risk Analysis, Statistics, Optimization and Modeling)

PROFESSIONAL EXPERIENCES

2023-2025

Teaching

MACHINE LEARNING & INFORMATICS LAB · Sorbonne University & Polytech Sorbonne



Labs for M1 DAC student in machine learning (Sorbonne university), Python (3rd year students at Polytech) and industrial Project (4th year students at Polytech).

2022

Research internship

AI FOR SCIENCE: PHYSICS BASED DEEP LEARNING FOR MODELING COMPLEX DYNAMICS. APPLICATION TO CLIMATE · ISIR (MLIA) bibliography, evaluation of using parcimonious implicit neural representations to solve partial differential equations.



2021

End of studies intership

COLOR TRANSFER FOR IMAGE EDITING · DxO Labs Developpement of an algorithms to compute color transfer between photography (Research in python and prototyping in C++)



2020

Ingeniering intership

OPTIMIZATION OF DATA PROCESSING FROM A LASER VELOCIMETER · IDIL fibres optiques Modeling of the operation of a fiber VISAR, analysis of the influence of parameters on the outputs, noise treatment



OTHERS

Photography

Landscapes, sport, astronomy...

Canon EOS 100D, Canon EOS R7.

Sport

Track and fields, Triathlon ...

Daily trainings.

Music

Piano



Informations

Lise Le Boudec
Lannion, France
Driving licence B & BE

Specialization

Deep Learning
Physics-informed Machine Learning
Neural Operator
Deep Learning for physical systems
Statistics
Optimization
Modelization

Interests

Deep Learning
Meteorology
Climate
Oceanography
Astronomy
Image processing

Languages

French | C2 (Native)
English | C1 (TOEIC 965/990)
German | B1
Italian | B1

Informatics

LaTeX, office
Python (Pytorch, OpenCV, Numpy, Matplotlib, Conda...)
Git
R, matlab
Java, C, C++
HTML, CSS
SQL, OWL (notion)

Social Networks

in Lise Le Boudec
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